

Homework 1: Due: Sept 18, 2026

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1.1 Point Processing

For this section, we will use the following notation:

- the pixel intensity of image I at coordinates (x, y) is denoted by $I[x, y]$
- a point-wise image processing operation is represented by the function $f(x, y)$
- the output of a point processing operation is denoted by the resulting image $\hat{I}$

Using this notation, write the equations for point processing operations described below:

1. Setting all pixels with intensity less than 50 to 0

Answer: Let S be the function that set all pixels with intensity less than 50 to 0

$$S(I) = \begin{cases} 0, & 0 < I < 50 \\ I & \text{otherwise} \end{cases}$$

2. Adding pixel-wise Gaussian noise with mean μ and standard deviation σ to the image

Answer:

Let G be the gaussian noise that produced with mean μ and standard deviation σ .

$$G(x, y) \sim \mathcal{N}(\mu, \sigma^2).$$

$$I'(x, y) = I(x, y) + G(x, y).$$

3. Gradual darkening an image from left to right

Answer: Let W be the with of the image

$$I(x, y) = I(x, y) \left(1 - \frac{x}{W - 1} \right)$$

4. Reducing the range of image intensities from $[0, 255]$ to $[50, 150]$

Answer:

$$I_{\text{new}} = \left(\frac{I_{\text{old}}}{255} \right) \times 100 + 50$$

5. “Quantizing” the image to n fixed values, i.e. instead of 256 possible pixel intensities between $[0, 255]$, we will only use n possible values between $[0, 255]$. What point processing equation would you use to achieve this? Write a general equation for any n .

(Hint: think about what you would do for $n = 2, 3, 4$ and then come up with a general formula)

Answer:

Let $n = 2, 3, 4$ be the quantized level. For each level, we want to split the range into n range. This can be done by normalization and scale.

To normalize, first we divided the original intensity with 255. Then, we can scale it by level $n - 1$.

The scaled value formula is

$$I = \frac{I}{255} \times (n - 1)$$

Since we are dealing with discrete value, we round this scaled value to a whole number. Round could be floor or ceil.

Finally, we scaled it down to the original bucket index.

$$I_{new} = round\left(\frac{I_{scaled}}{255}\right) \times (n - 1) \times \frac{255}{n - 1}$$

1.2 Lenses

1. Draw a ray diagram showing an object, a convex lens, and the image plane. Show the rays if the object is at infinity. Show the focal length of the lens in the ray diagram. You can make these drawings either by hand or with your computer (e.g., PowerPoint, Paint, or any other diagram tools).

Answer:

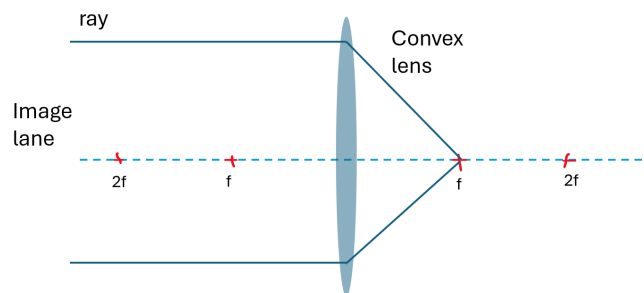


Figure 1.1: Ray diagram

2. You have a pinhole camera. Abraham Lincoln, a person with height 200 cm, goes and stands 1 m in front of the camera. If Lincoln demands that the image should have $20\times$ magnification, how far should the image plane be from the pinhole?

Answer: The $20\times$ magnification can be calculated by

$$\frac{s_i}{s_0} = 20$$

Since $s_0 = 1m$,

$$\frac{s_i}{1} = 20$$

Thus, $s_i = 20m$. The image plane should be 20m away from the pinhole.

3. You now have a new camera with a lens. Lincoln demands that you must photograph his hat, which has a height and width of 25 cm. You have a lens with focal length 35 mm, and the image sensor is placed at the focus. The image sensor width and height is 50 mm.

- (a) What is the minimum distance between Lincoln's hat and the camera for the entire hat to fit within the photo?

Answer:

We have the formula

$$\frac{y_i}{y_0} = \frac{s_i}{s_0}$$

$$\frac{50}{250} = \frac{35}{s_0}$$

Therefore, $s_0 = 175mm$ which is the minimum distance between Lincoln's hat and the camera for the entire hat to fit within the photo.

- (b) What is the distance if Lincoln demands that the hat should only occupy quarter of the area of the image?

Answer:

We have the formula

$$\frac{y_i}{y_0} = \frac{s_i}{s_0}$$

$$\frac{25}{250} = \frac{35}{s_0}$$

Therefore, $s_0 = 350mm$ which is the minimum distance between Lincoln's hat and the camera for the entire hat to fit within the photo.

1.3 Basic Operations in OpenCV

Code on in HW1.py. Comments are provided below.

Task 2.2: The image color displayed by matplotlib is different from the original image.

Task 3.3.c: Sum all the pixels in the difference image for the two methods, and report the number

- Nearest neighbor: 46599821
- Bicubic: 40261748

Bicubic caused less error in upsampling.

1.4 Edge Detection and Image Blurring by Convolution

All tasks on in HW1.py

1.5 Extra Credit: Images as Matrices

Q1: In class, we studied how to represent images as a matrix. Is every matrix an image? You can provide answers for or against this argument or argue on both sides.

Answer: No, every matrix is not an *image*. Matrix is just the data. But, every matrix can be converted to an image depending on the rules, e.g the format interpretation.

Q2. I have a camera that produces color images with 1000 rows and 1000 columns. I store pixel intensities as integers in the range $[0, 255]$. How many bytes of storage will I need to store this image? If I store this image in my computer as .jpeg, it will occupy approximately 500 kilobytes. Is this different from your answer? Why?

Answer: Since it's a color image (3 channels) of 1 millions pixels, it would take 3 million bytes to store this image. This is different from the 500 kilobytes stored as .jpeg. This is because .jpeg is a 'method of lossy compression for digital images'. The original image when saved as .jpeg has been compressed to help reduce the size while also reducing the image quality as a trade off.